

SignBridge AI

AI-Powered Assistive Smart Glasses for Two-Way Communication

Technical Project Report & Prototype Documentation

Project Type: AI + Computer Vision + IoT + Wearable Technology

Core Objective:

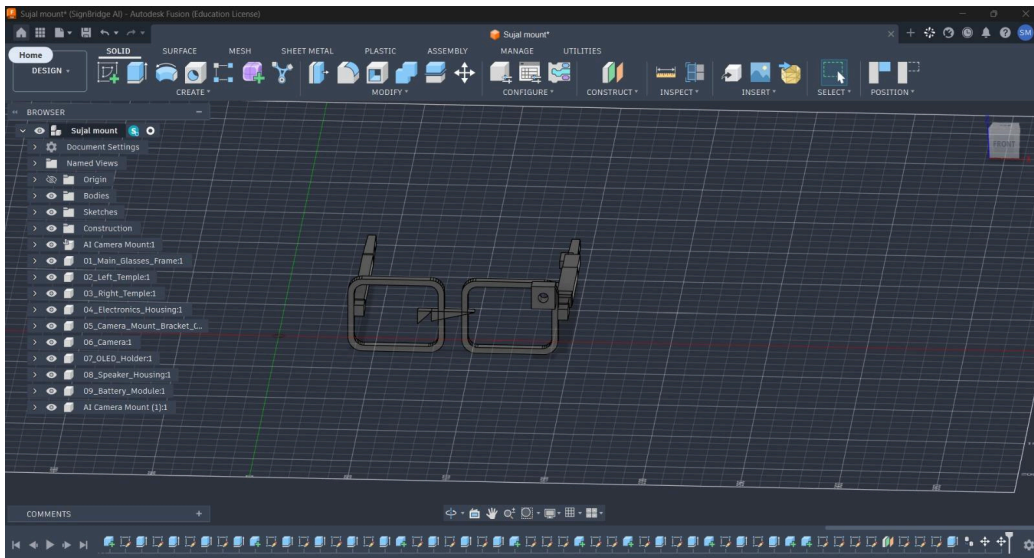
To develop a wearable smart-glasses system that enables two-way communication between sign-language users and people who do not understand sign language.

Key Technologies

- Artificial Intelligence
- Computer Vision
- Hand Landmark Detection
- Sign Language Recognition
- Speech-to-Text
- Text-to-Speech
- OLED Display
- Embedded Systems
- IoT / Wireless Connectivity

Team Name: Code Cr@ft

SIH: Smart India Hackathon 2026



1. Abstract

Communication between deaf or non-speaking individuals and people unfamiliar with sign language can be difficult in everyday situations. Existing communication methods often require a third person, smartphone interaction, or manual typing, which can reduce the speed and naturalness of communication.

SignBridge AI proposes an AI-powered assistive smart-glasses system designed to provide a wearable interface for two-way communication.

The system uses a camera to capture sign-language gestures and an AI-based computer-vision pipeline to process and recognize the gestures. The recognized sign is converted into text and then into speech, allowing the other person to hear the intended message.

In the reverse direction, speech from the other person is captured through a microphone, converted into text using Speech-to-Text technology, and displayed to the wearer through an OLED display.

The overall objective is to create a compact, hands-free and scalable communication system that combines AI with wearable hardware.

2. Problem Statement

People who communicate primarily through sign language may face communication barriers when interacting with individuals who do not understand sign language.

Major challenges:

- Lack of common communication language.
- Dependence on interpreters in many situations.
- Typing messages manually can be slow.
- Smartphone-based communication is not always convenient.
- Real-time sign-language recognition is technically challenging.
- Wearable integration requires compact hardware and efficient processing.

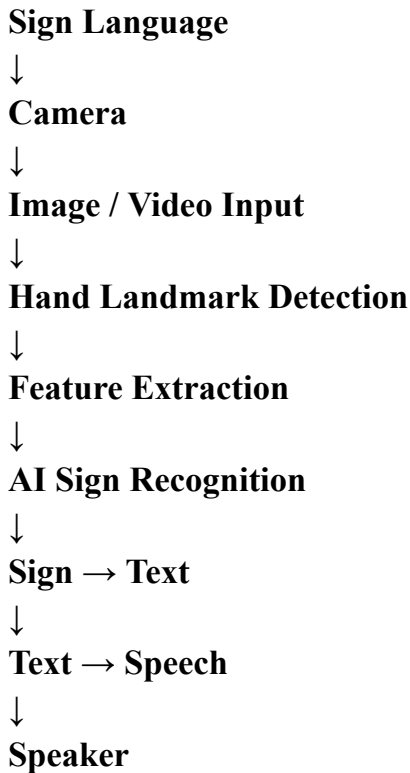
Therefore, there is a need for a practical system capable of supporting **real-time, two-way communication through a wearable interface**.

3. Proposed Solution

SignBridge AI is designed as a pair of smart glasses containing or connected to the necessary sensing, processing and communication components.

The system provides two primary communication directions.

Direction 1: Sign Language → Speech





Audio Output

The camera captures the user's hand gestures. Computer-vision techniques identify relevant hand landmarks and extract features. The AI recognition system then determines the corresponding sign and converts it into text. The text is subsequently converted into speech.

Direction 2: Speech → Text

Speech



Microphone



Speech-to-Text



Text Processing



OLED Display

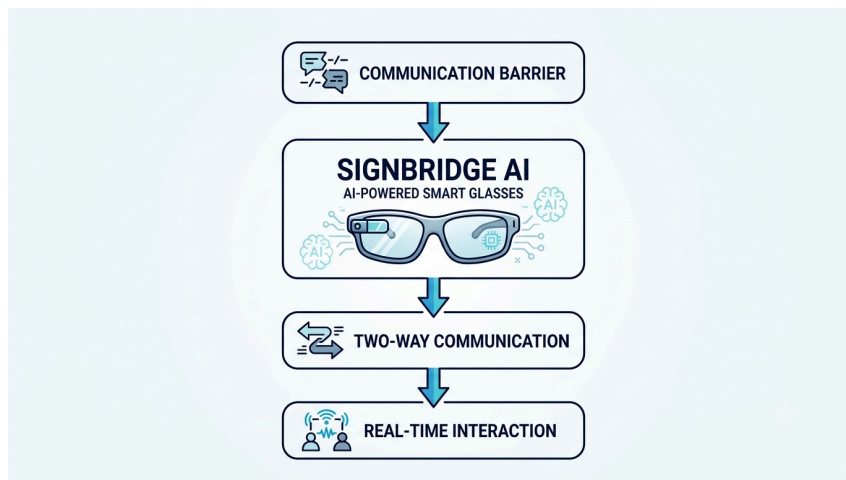


User Reads Text

The other person can speak normally. The microphone captures the voice, which is converted into text. The resulting text is displayed to the sign-language user through the wearable display.

Core Concept :

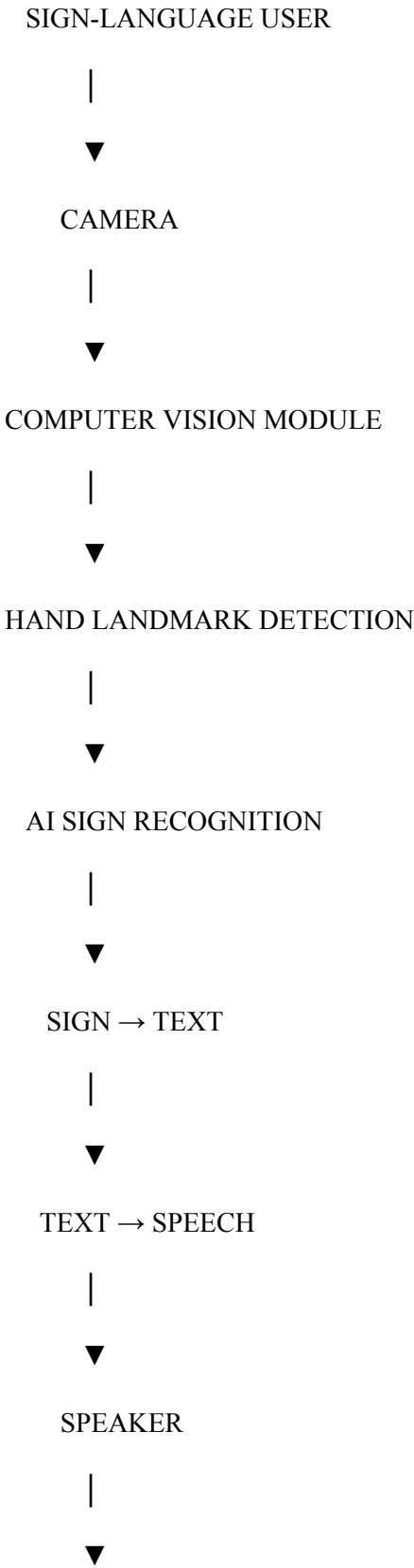
One wearable interface. Two communication directions.



4. System Architecture

The SignBridge AI architecture consists of interconnected hardware and software modules.

High-Level Architecture



Reverse Communication

OTHER PERSON



MICROPHONE



SPEECH-TO-TEXT



TEXT PROCESSING



OLED DISPLAY



SIGN-LANGUAGE USER

Main System Components

Component	Purpose
Camera	Captures hand gestures
AI/CV Module	Processes and recognizes signs
Microphone	Captures speech
Speech-to-Text	Converts speech into text
OLED display	Displays received text
Speaker	Produces converted audio
Processing Unit	Executes software/AI operations

Battery	Provides portable power
Wireless Connectivity	Enables communication between modules where required

5. Smart Glasses Design

The physical design focuses on integrating electronic components into a wearable glasses frame while maintaining a practical form factor.

Design Considerations

- Compact component placement
- Camera positioning
- Display visibility
- Speaker placement
- Microphone positioning
- Battery placement
- Wiring and connectivity
- User comfort
- Modularity
- Ease of assembly

Hardware Integration

The glasses frame acts as the physical platform for integrating the different modules.

Camera:

Positioned to capture the user's hand gestures within the required field of view.

OLED Display:

Provides visual text output to the wearer.

Speaker:

Provides audio output for recognized signs.

Microphone:

Captures speech from the person communicating with the wearer.

Processing Unit:

Handles communication between the hardware and software components and supports AI processing depending on the selected implementation.

Battery:

Provides portable power to the wearable electronics.

Mechanical Design

The project includes CAD-based development for the glasses structure and component mounting.

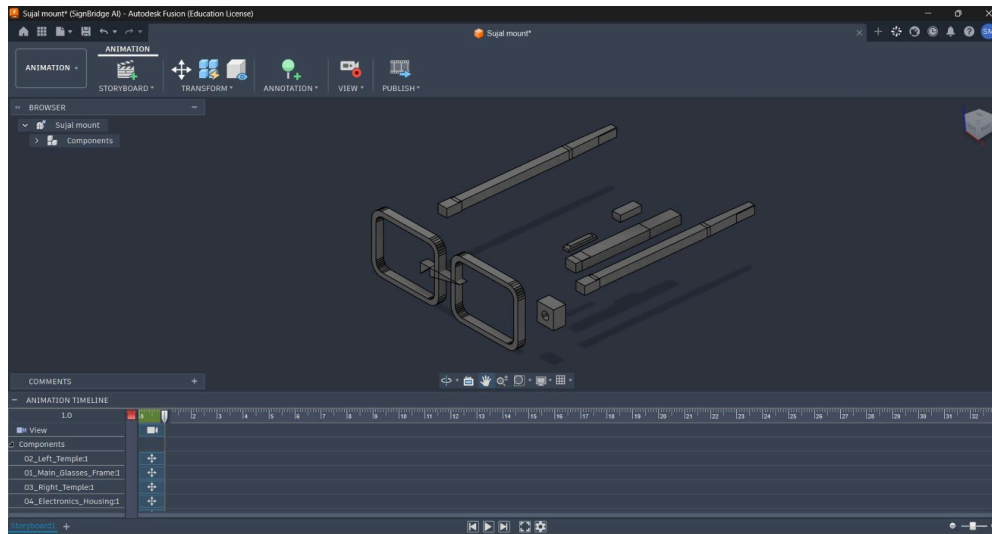


Figure 5.1: Smart glasses CAD design

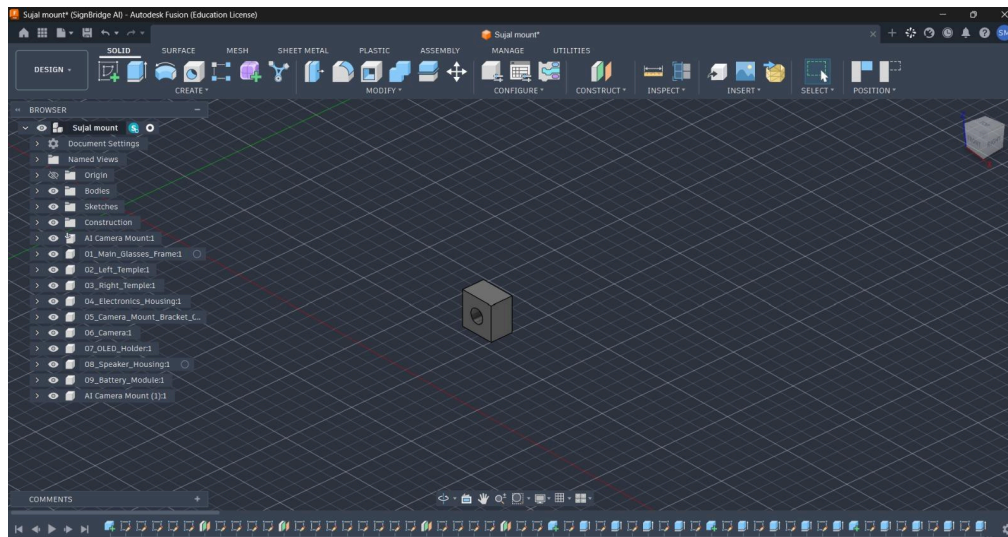


Figure 5.2: Camera mounting design

6. Mechanical Development and Prototyping

The project follows a digital-to-physical development workflow.

Development Workflow

CONCEPT



CAD MODEL



COMPONENT DESIGN



CAM SETUP



TOOLPATH GENERATION



G-CODE



MANUFACTURING / FABRICATION



HARDWARE ASSEMBLY



WORKING PROTOTYPE

CAD Development

Computer-Aided Design was used to develop the physical structure and component mounting arrangement.

The CAD stage helps in:

- Designing the glasses frame.
- Planning component placement.
- Designing the camera mount.
- Checking physical integration.
- Iterating the mechanical design before fabrication.

CAM & G-Code

The manufacturing workflow involves converting the designed geometry into machine-readable instructions where applicable.

CAD Model → CAM Setup → Toolpath → G-Code

G-code provides instructions required by compatible manufacturing equipment for producing the designed component.

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M8
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G1 Z-8.91 F333.3
X7.519 Y-5.018 Z-8.944 F1000
X7.5 Y-5.015 Z-8.974
X7.471 Y-5.011 Z-8.993
X7.437 Y-5.005 Z-9
X7.412 Y-5.001
X7.397 Y-5
X-2.5
X-7.391
X-7.486 Y-5.001
X-7.491 Y-5.015
X-7.529 Y-5.021 Z-8.992
X-7.561 Y-5.025 Z-8.971
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G0 X-6.1 Y4.8
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Figure 6.1: G-Code

Prototype Development

The project is currently being developed through a combination of mechanical design, component integration and software development.

The development workflow is:

Concept → CAD Design → Component Design → Manufacturing → Hardware Integration → AI/Software Integration

The current work demonstrates the **mechanical and manufacturing aspects of the proposed wearable system**, while further integration of the AI, sensing and communication modules remains part of the ongoing development.

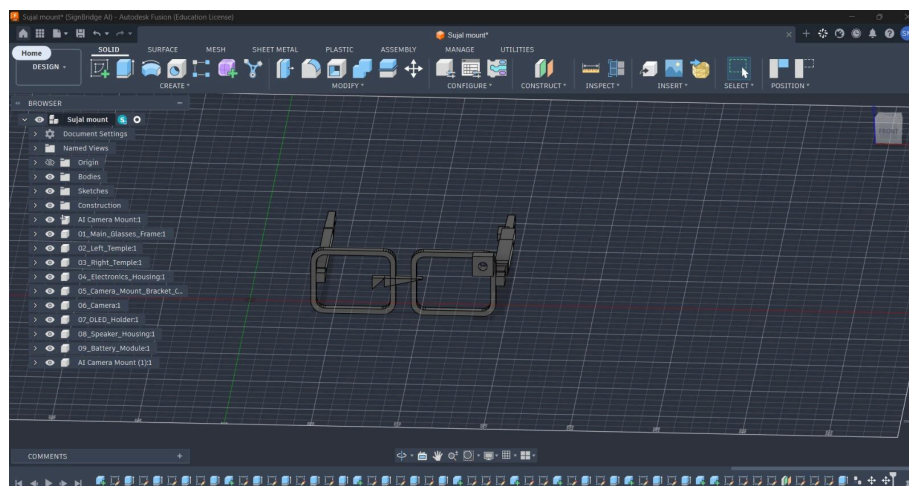
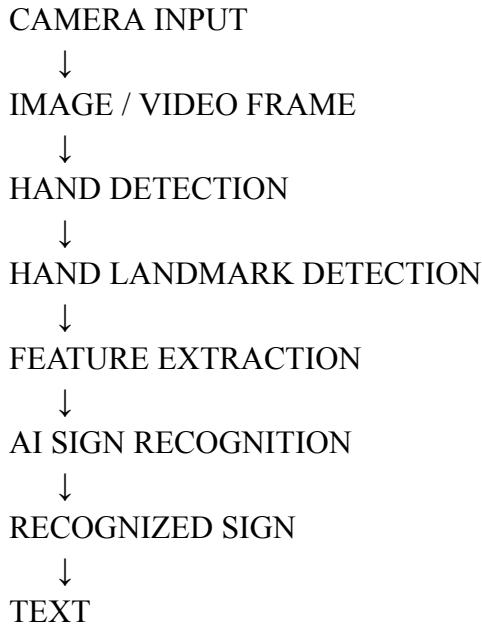


Figure 6.2 : CAD model of the proposed smart-glasses frame

7. AI and Computer Vision Pipeline

Computer vision forms the core of the sign-recognition side of SignBridge AI.

Processing Pipeline



Hand Landmark Detection

Instead of relying only on raw images, hand landmarks can represent important points on the hand, such as finger joints and other key locations.

These landmarks provide structured information about hand position and configuration.

Feature Extraction

The extracted hand information is converted into features that can be used by the recognition model.

Depending on the final implementation, features may include:

- Landmark positions
- Relative distances
- Hand orientation
- Finger configuration
- Movement information for dynamic signs

AI Sign Recognition

The recognition model processes the extracted features and determines the corresponding sign.

The recognized output is then passed to the text-processing stage.

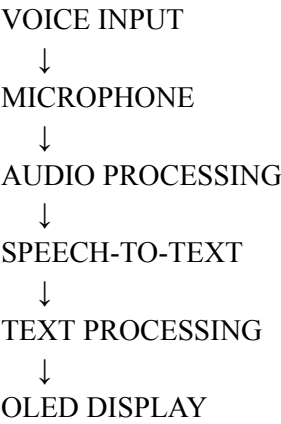
Current Development Status

Module	Status
Smart-glasses mechanical design	Designed
Camera mounting	Designed
CAD development	Completed/Demonstrated
CAM / G-code workflow	Demonstrated
Sign recognition pipeline	Under development
Speech-to-text pipeline	Proposed / Under development
Wearable integration	Future development

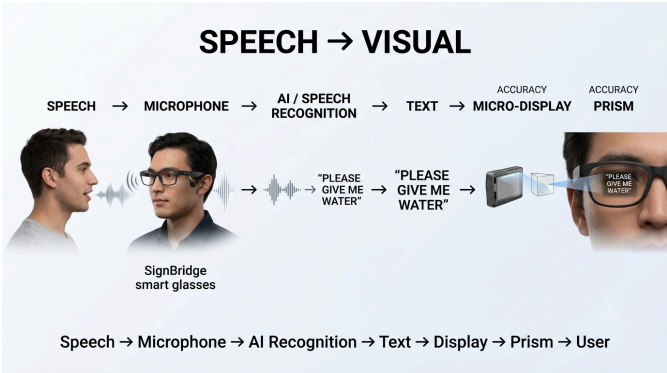
8. Speech Processing

The reverse communication pipeline allows spoken communication to be converted into readable text.

Speech-to-Text Pipeline

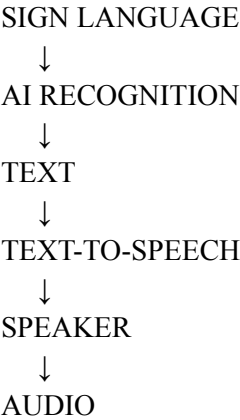


The microphone captures the speaker's voice. A Speech-to-Text system converts the audio into textual information, which is then displayed to the wearer.



Text-to-Speech Pipeline

For the opposite direction:

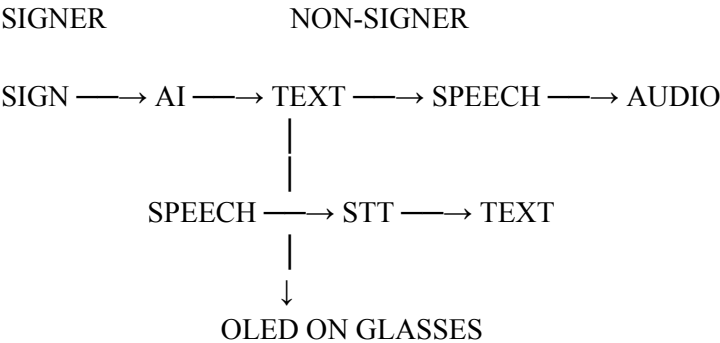


The recognized sign is converted into text and subsequently synthesized into speech.

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Two-Way Communication

The two pipelines work independently but form one communication system.



This creates a **bidirectional communication interface** without requiring the wearer to continuously type messages.

The current prototype demonstrates the project direction through camera-based sign input and recognition experiments.

9. Feasibility and Applications

Technical Feasibility

The project combines established technologies including:

- Computer vision
- Hand landmark detection
- AI-based classification
- Speech-to-Text
- Text-to-Speech
- Microcontrollers / processing devices
- OLED displays
- Cameras and microphones

These technologies provide a foundation for developing the proposed system.

Manufacturing Feasibility

The use of CAD, CAM and fabrication workflows allows the physical design to be iterated and prototyped.

Scalability

The system can potentially be expanded through:

- Larger sign vocabulary
 - Improved recognition models
 - Continuous sign recognition
 - Multilingual speech
-
- Improved hardware integration
 - Smaller electronics
 - Better power management
 - On-device AI processing

Applications

Education

Communication support between students and teachers.

Healthcare

Communication assistance between patients and healthcare professionals.

Workplace

Improved interaction between employees and colleagues.

Public Services

Communication assistance in banks, offices, transport facilities and government services.

Daily Life

Everyday conversations with people unfamiliar with sign language.

Social Interaction

Reducing communication barriers during regular social interactions.

Current Limitations

The project still faces several technical challenges:

- Limited sign vocabulary during early development.
- Recognition accuracy can vary depending on the gesture and environment.
- Continuous sign-language recognition is more complex than recognizing isolated signs.
- Lighting conditions can affect camera-based recognition.
- Background objects may interfere with hand detection.
- Real-time processing may introduce latency.
- Wearable hardware has size and power constraints.
- Miniaturizing all components into a comfortable glasses frame is challenging.

10. Future Development

The planned development roadmap is:

CURRENT PROTOTYPE



IMPROVED AI MODEL



LARGER SIGN VOCABULARY



CONTINUOUS SIGN RECOGNITION



REAL-TIME OPTIMIZATION



ON-DEVICE AI



HARDWARE MINIATURIZATION



FULLY INTEGRATED WEARABLE

Future Improvements

1. Expand the number of recognizable signs.
2. Support continuous sign-language sentences.
3. Improve recognition accuracy.
4. Reduce processing latency.
5. Improve performance under different lighting conditions.
6. Integrate AI processing more efficiently.
7. Improve battery efficiency.
8. Further miniaturize electronic components.
9. Improve glasses ergonomics and comfort.
10. Add multilingual speech support.

Research Gap

Existing research demonstrates individual capabilities such as sign-language recognition, speech processing and assistive communication. However, integrating these technologies into a **practical wearable system supporting two-way communication** introduces additional challenges involving:

- Real-time processing
- Hardware miniaturization
- Power consumption
- Wearability
- Recognition reliability
- Integration of multiple communication modules

SignBridge AI focuses on bringing these elements together into one wearable communication platform.

Conclusion

SignBridge AI proposes an AI-powered smart-glasses system for assisting two-way communication between sign-language users and non-signers.

The proposed system combines **computer vision, AI-based sign recognition, Speech-to-Text, Text-to-Speech and wearable hardware** into a unified communication platform.

The project has progressed through stages of **CAD design, camera-mount development, manufacturing workflow and prototype development**, providing a foundation for further AI and hardware integration.

The long-term objective is to develop a compact, practical and scalable wearable system that can help make everyday communication more accessible.

SignBridge AI

SIGN → AI → SPEECH
SPEECH → TEXT → SIGNER

“Making communication more accessible through AI-powered wearable technology.”